



AMITY UNIVERSITY PUNJAB, MOHALI

B.SC. MATHEMATICS (5TH SEMESTER)
SET THEORY AND METRIC SPACES (MAT304)

The Cantor Set

A Comprehensive Study

Dr. Lokesh Singh

`lsingh@pb.amity.edu`

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1 Introduction to the Cantor Set

1.1 Historical Context

The Cantor set, discovered by the German mathematician **Georg Cantor** in 1883, is one of the most remarkable and counterintuitive objects in all of mathematics. It appears in Cantor's paper "*De la puissance des ensembles parfaits de points*" (On the power of perfect sets of points).

Cantor constructed this set to demonstrate that there exist **perfect** sets (closed sets with no isolated points) that are **nowhere dense** yet **uncountable**. Before this discovery, mathematicians had assumed that any uncountable set must contain an interval. The Cantor set shattered this intuition.

Remark 1.1. The Cantor set is often called **Cantor's Dust** because it consists of infinitely many points scattered like dust, yet it has the same cardinality as the entire real line!

2 Construction of the Cantor Set

2.1 The Geometric Construction (Middle Thirds)

The Cantor set is constructed by repeatedly removing the open middle third of each remaining interval. This is the most intuitive way to visualize it.

Definition 2.1 (Cantor Set via Iterative Removal). Start with the closed unit interval $C_0 = [0, 1]$. For each step $n \geq 1$, remove the open middle third of each interval in C_{n-1} to obtain C_n . The Cantor set $\mathcal{C}$ is defined as:

$$\mathcal{C} = \bigcap_{n=0}^{\infty} C_n.$$

Example 2.1 (First Few Iterations). • **Step 0:** $C_0 = [0, 1]$

- **Step 1:** Remove $(\frac{1}{3}, \frac{2}{3})$.

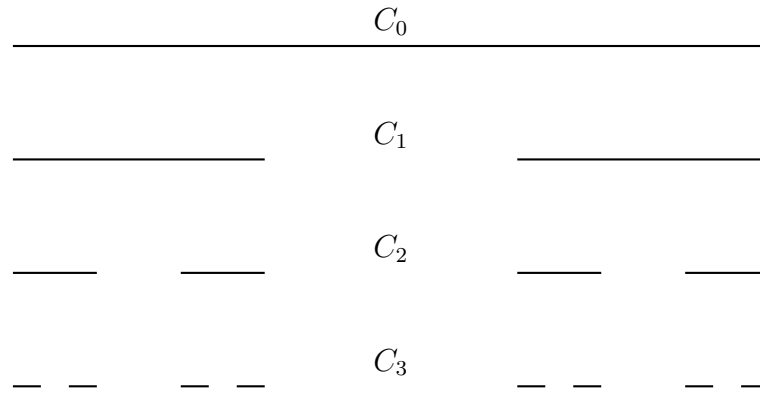
$$C_1 = \left[0, \frac{1}{3}\right] \cup \left[\frac{2}{3}, 1\right].$$

- **Step 2:** Remove middle thirds of each interval:

$$C_2 = \left[0, \frac{1}{9}\right] \cup \left[\frac{2}{9}, \frac{1}{3}\right] \cup \left[\frac{2}{3}, \frac{7}{9}\right] \cup \left[\frac{8}{9}, 1\right].$$

- **Step 3:** Remove middle thirds of each of the four intervals:

$$C_3 = \left[0, \frac{1}{27}\right] \cup \left[\frac{2}{27}, \frac{1}{9}\right] \cup \left[\frac{2}{9}, \frac{7}{27}\right] \cup \left[\frac{8}{27}, \frac{1}{3}\right] \cup \left[\frac{2}{3}, \frac{19}{27}\right] \cup \left[\frac{20}{27}, \frac{7}{9}\right] \cup \left[\frac{8}{9}, \frac{25}{27}\right] \cup \left[\frac{26}{27}, 1\right].$$



2.2 The Ternary Representation

The more effective way to understand the Cantor set is through its ternary (base-3) expansion.

Definition 2.2 (Ternary Expansion). Every number $x \in [0, 1]$ can be written uniquely as:

$$x = \sum_{n=1}^{\infty} \frac{a_n}{3^n}, \quad a_n \in \{0, 1, 2\},$$

except for the countably many numbers with terminating ternary expansions (which end in infinitely many 2's).

Theorem 2.2 (Ternary Characterization).

$$\mathcal{C} = \left\{ x \in [0, 1] : x = \sum_{n=1}^{\infty} \frac{a_n}{3^n}, \text{ where } a_n \in \{0, 2\} \text{ for all } n \right\}.$$

In words: **The Cantor set consists exactly of those numbers in $[0, 1]$ whose ternary expansion contains only 0's and 2's.**

Proof. At step n , the intervals removed are precisely those numbers whose n -th ternary digit is 1. For example:

- At step 1, we remove $(\frac{1}{3}, \frac{2}{3})$, which is all numbers with $a_1 = 1$.
- At step 2, we remove $(\frac{1}{9}, \frac{2}{9})$ (numbers with $a_1 = 0, a_2 = 1$) and $(\frac{7}{9}, \frac{8}{9})$ (numbers with $a_1 = 2, a_2 = 1$).

Therefore, a number survives all removals iff none of its ternary digits are equal to 1. Since we use the non-terminating ternary expansion (avoiding trailing 2's), this is equivalent to all digits being 0 or 2. $\square$

Example 2.3. The following numbers are in the Cantor set:

- $0 = 0.000 \dots_3$
- $1 = 0.222 \dots_3$
- $\frac{1}{3} = 0.02222 \dots_3 = 0.1000 \dots_3$ (using the non-terminating expansion, it is in $\mathcal{C}$)
- $\frac{2}{3} = 0.2000 \dots_3 = 0.1222 \dots_3$ (in $\mathcal{C}$)
- $\frac{1}{4} = 0.020202 \dots_3$ (in $\mathcal{C}$)
- $\frac{1}{2} = 0.1111 \dots_3$ (not in $\mathcal{C}$)

3 Structural Properties of the Cantor Set

3.1 Cardinality: $\mathcal{C}$ is Uncountable

Theorem 3.1. The Cantor set $\mathcal{C}$ has cardinality $\mathfrak{c} = 2^{\aleph_0}$.

Proof. We construct a bijection between $\mathcal{C}$ and $\{0, 1\}^{\mathbb{N}}$. Given $x \in \mathcal{C}$, its ternary expansion has digits $a_n \in \{0, 2\}$. Define:

$$\phi(x) = (b_1, b_2, b_3, \dots), \quad \text{where } b_n = \begin{cases} 0 & \text{if } a_n = 0, \\ 1 & \text{if } a_n = 2. \end{cases}$$

This is a bijection (with the usual care taken for the countably many numbers with two ternary expansions). Therefore:

$$|\mathcal{C}| = |\{0, 1\}^{\mathbb{N}}| = 2^{\aleph_0} = \mathfrak{c}.$$

□

Corollary 3.2. The Cantor set has the same cardinality as $\mathbb{R}$.

Remark 3.3 (Profound Implication). This is deeply counterintuitive: $\mathcal{C}$ is uncountable, yet it contains **no intervals** and has **no interior points**. It shows that "uncountable" does not imply "large" in a geometric sense.

3.2 Length (Measure): $\mathcal{C}$ has Measure Zero

Theorem 3.4. The Cantor set has Lebesgue measure zero.

Proof. At step n , the set C_n consists of 2^n intervals, each of length $1/3^n$. Therefore:

$$\text{length}(C_n) = 2^n \cdot \left(\frac{1}{3^n}\right) = \left(\frac{2}{3}\right)^n.$$

Since $\mathcal{C} \subset C_n$ for all n , we have:

$$0 \leq \text{length}(\mathcal{C}) \leq \text{length}(C_n) = \left(\frac{2}{3}\right)^n \rightarrow 0.$$

By the Squeeze Theorem, $\text{length}(\mathcal{C}) = 0$.

□

Remark 3.5. This is a classic counterexample: an uncountable set can have measure zero. Before Cantor, mathematicians believed that uncountable sets must have positive length.

3.3 Nowhere Dense

Definition 3.1. A set A is **nowhere dense** if its closure has empty interior. Equivalently, the closure of A contains no open interval.

Theorem 3.6. The Cantor set is nowhere dense in $\mathbb{R}$.

Proof. Since $\mathcal{C}$ is closed (we will prove this next), $\overline{\mathcal{C}} = \mathcal{C}$. If $\mathcal{C}$ contained an open interval, that interval would have positive length. But $\mathcal{C}$ has measure zero. Therefore $\mathcal{C}$ contains no interval. Hence $\mathcal{C}$ has empty interior and is nowhere dense.

□

3.4 Closed and Compact

Theorem 3.7. The Cantor set $\mathcal{C}$ is closed.

Proof. Each C_n is a finite union of closed intervals, hence closed. The intersection of closed sets is closed. Therefore:

$$\mathcal{C} = \bigcap_{n=0}^{\infty} C_n$$

is closed. □

Theorem 3.8. The Cantor set $\mathcal{C}$ is compact.

Proof. $\mathcal{C} \subset [0, 1]$ is bounded and closed. By the Heine-Borel theorem, it is compact. □

Remark 3.9. Compactness is a key property used in many applications. Being compact means every open cover has a finite subcover, and every sequence has a convergent subsequence.

3.5 Perfect (Closed with No Isolated Points)

Definition 3.2. A set is **perfect** if it is closed and every point is a limit point (i.e., has no isolated points).

Theorem 3.10. The Cantor set is perfect.

Proof. We already know $\mathcal{C}$ is closed. Now take any $x \in \mathcal{C}$ and any $\varepsilon > 0$. We must find $y \in \mathcal{C}$, $y \neq x$, such that $|x - y| < \varepsilon$.

Write $x = \sum_{n=1}^{\infty} a_n/3^n$ with $a_n \in \{0, 2\}$. Choose N large enough so that $3^{-N} < \varepsilon$. Flip the N -th digit:

$$b_n = \begin{cases} a_n & \text{if } n \neq N, \\ 2 - a_n & \text{if } n = N. \end{cases}$$

(So if $a_N = 0$, $b_N = 2$; if $a_N = 2$, $b_N = 0$.) Let $y = \sum b_n/3^n$. Then $y \in \mathcal{C}$, $y \neq x$, and:

$$|x - y| = \frac{|a_N - b_N|}{3^N} = \frac{2}{3^N} < \varepsilon.$$

Thus every point of $\mathcal{C}$ is a limit point. Hence $\mathcal{C}$ is perfect. □

3.6 Totally Disconnected

Definition 3.3. A topological space is **totally disconnected** if its only connected subsets are singletons (and the empty set).

Theorem 3.11. The Cantor set is totally disconnected.

Proof. Let $x < y$ be two points in $\mathcal{C}$. Since the gaps between intervals in the construction are open, there exists a removed interval (a, b) such that $x < a < b < y$. Then:

$$\mathcal{C} \cap (-\infty, a) \quad \text{and} \quad \mathcal{C} \cap (b, \infty)$$

are disjoint, non-empty, relatively open subsets of $\mathcal{C}$ whose union is $\mathcal{C}$ (minus the gap). Hence $\mathcal{C}$ is disconnected. Since this holds for any two points, $\mathcal{C}$ has no intervals, and its connected components are singletons. □

3.7 Homeomorphic to the Cantor Space

Definition 3.4. The **Cantor space** is the set $\{0, 1\}^{\mathbb{N}}$ with the product topology (where $\{0, 1\}$ is discrete).

Theorem 3.12. $\mathcal{C}$ is homeomorphic to $\{0, 1\}^{\mathbb{N}}$.

Proof. The map $\phi : \mathcal{C} \rightarrow \{0, 1\}^{\mathbb{N}}$ defined by the ternary digit map ($0 \rightarrow 0, 2 \rightarrow 1$) is a homeomorphism. It is continuous, bijective, and has a continuous inverse. $\square$

Remark 3.13. This is extremely important: it shows that the Cantor set is a **universal** object. Every compact metric space is a continuous image of the Cantor set (the **Cantor-Hausdorff theorem**).

4 Properties as a Subset of $\mathbb{R}$

4.1 $\mathcal{C}$ Contains No Intervals

We have already established this via measure zero and nowhere dense. But we can also prove it directly: if $\mathcal{C}$ contained an interval, that interval would contain a number whose ternary expansion has a 1 somewhere—contradiction.

4.2 $\mathcal{C}$ is Symmetric

The construction is symmetric about $\frac{1}{2}$. If $x \in \mathcal{C}$, then $1 - x \in \mathcal{C}$.

4.3 $\mathcal{C}$ is Self-Similar

The Cantor set is a fractal. It is the unique non-empty compact set satisfying:

$$\mathcal{C} = \frac{1}{3}\mathcal{C} \cup \left(\frac{2}{3} + \frac{1}{3}\mathcal{C}\right).$$

This is the basis of the **Iterated Function System** (IFS) construction.

4.4 $\mathcal{C}$ is the Boundary of the Cantor Set Itself

A curious property: $\mathcal{C} = \partial\mathcal{C}$ (every point is a boundary point).

4.5 $\mathcal{C}$ is a Metric Space

As a subspace of $\mathbb{R}$, $\mathcal{C}$ is a metric space with the usual distance $d(x, y) = |x - y|$. It is:

- **Complete:** Since it is closed in $\mathbb{R}$.
- **Compact:** Since it is closed and bounded.
- **Separable:** It has a countable dense subset (e.g., endpoints of removed intervals).

5 Cantor Set Properties (Metric Spaces)

5.1 Completeness

$\mathcal{C}$ is complete as a closed subset of the complete space $\mathbb{R}$.

5.2 Compactness

$\mathcal{C}$ is compact. This is a standard example of a compact set that is not an interval.

5.3 Separability

$\mathcal{C}$ is separable. The set of endpoints of removed intervals is countable and dense in $\mathcal{C}$.

5.4 Perfect Sets

The Cantor set is a perfect set. The Cantor-Bendixson theorem states that every closed set can be written as the union of a perfect set and a countable set.

5.5 Baire Category

Although $\mathcal{C}$ is nowhere dense, it is a Baire space (since it is complete).

5.6 Continuous Functions

There exists a continuous surjection from $\mathcal{C}$ onto $[0, 1]$ (the Cantor function, also called the Devil's Staircase). This function is continuous, non-decreasing, and has derivative zero almost everywhere.

6 Related Constructions

6.1 Fat Cantor Sets (Smith-Volterra-Cantor Sets)

Instead of removing middle thirds, remove middle fractions whose lengths sum to less than 1. The resulting set is:

- Nowhere dense
- Perfect
- Uncountable
- **Has positive Lebesgue measure!**

This shows that "nowhere dense" does not imply "measure zero."

6.2 Generalized Cantor Sets

For any sequence (a_n) with $0 < a_n < 1$, we can remove the middle a_n -th portion at step n . The resulting set has various properties depending on the sequence.

6.3 Cantor Dust in Higher Dimensions

The product $\mathcal{C} \times \mathcal{C}$ is a Cantor set in the plane. Higher-dimensional generalizations exist.

7 Exercises

1. Prove that every point of the Cantor set is a limit point.
2. Show that the Cantor set is uncountable by constructing an injection from $\{0, 1\}^{\mathbb{N}}$ into $\mathcal{C}$.
3. Prove that the Cantor set has Lebesgue measure zero.
4. Show that the Cantor set is nowhere dense.
5. Prove that the Cantor set is totally disconnected.
6. Show that the Cantor set is homeomorphic to $\{0, 1\}^{\mathbb{N}}$.
7. Construct the Fat Cantor set with positive measure and show it is nowhere dense.
8. Prove that $\mathcal{C} + \mathcal{C} = [0, 2]$, where $\mathcal{C} + \mathcal{C} = \{x + y : x, y \in \mathcal{C}\}$.
9. Show that the Cantor function is continuous but not absolutely continuous.
10. Prove that $\mathcal{C}$ is not countable using a diagonal argument directly on ternary expansions.

8 Summary: The Cantor Set at a Glance

Property	Status
Cardinality	$\mathfrak{c} = 2^{\aleph_0}$ (uncountable)
Lebesgue Measure	0
Nowhere Dense	Yes
Closed	Yes
Compact	Yes
Perfect	Yes (closed + no isolated points)
Totally Disconnected	Yes
Self-Similar	Yes (fractal dimension $\log_3 2$)
Homeomorphic to	$\{0, 1\}^{\mathbb{N}}$
Complete	Yes
Separable	Yes

The Cantor Set is a perfect, compact, nowhere dense, totally disconnected, uncountable subset of $\mathbb{R}$ with Lebesgue measure zero. It is the universal compact metric space in the sense that every compact metric space is a continuous image of it. It serves as a counterexample to many naive intuitions about uncountable sets and is foundational in set theory, topology, measure theory, and dynamical systems.

References

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